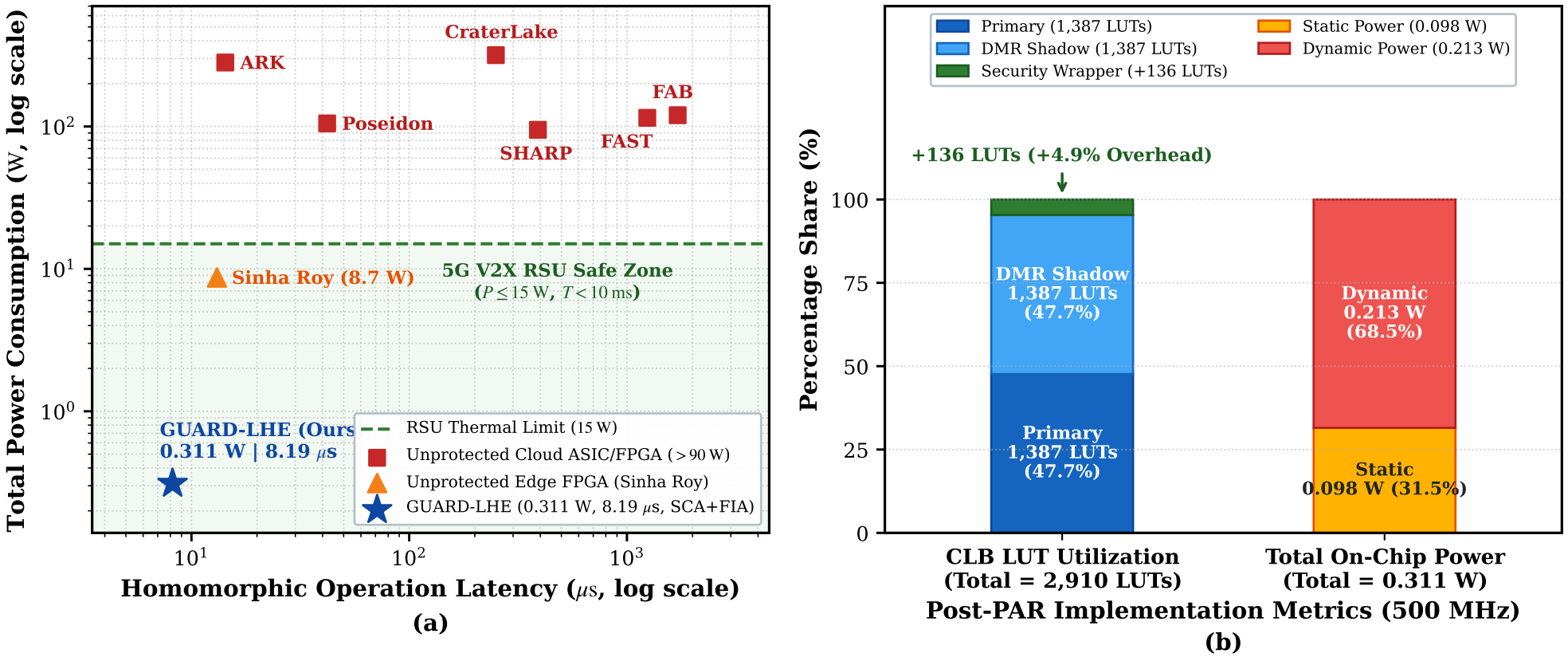
Total Power Consumption (W, log scale)
10^2
10^1
10^0
ARK
CraterLake
Poseidon
SHARP
FAST
FAB
Sinha Roy (8.7 W)
5G V2X RSU Safe Zone
($P \leq 15$ W, $T < 10$ ms)
GUARD-LHE (Ours
0.311 W | 8.19 μs
RSU Thermal Limit (15 W)
Unprotected Cloud ASIC/FPGA (>90 W)
Unprotected Edge FPGA (Sinha Roy)
GUARD-LHE (0.311 W, 8.19 μs, SCA+FIA)
10^1
10^2
10^3
Homomorphic Operation Latency (μs, log scale)
(a)
Primary (1,387 LUTs)
DMR Shadow (1,387 LUTs)
Security Wrapper (+136 LUTs)
Static Power (0.098 W)
Dynamic Power (0.213 W)
+136 LUTs (+4.9% Overhead)
Percentage Share (%)
100
75
50
25
0
DMR Shadow
1,387 LUTs
(47.7%)
Primary
1,387 LUTs
(47.7%)
Dynamic
0.213 W
(68.5%)
Static
0.098 W (31.5%)
CLB LUT Utilization
(Total = 2,910 LUTs)
Total On-Chip Power
(Total = 0.311 W)
Post-PAR Implementation Metrics (500 MHz)
(b)